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Free vibration analysis of nonsymmetrically laminated cross-ply L-shaped frames

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Abstract

Fiber-reinforced composite laminates can be tailored to produce structural elastic couplings that optimize their response in dynamic environments. It is therefore essential that the free vibration characteristics of structural elements fabricated from fiber-reinforced composites be accurately modeled and investigated. In this work, an analytical wave-based approach is extended to study the in-plane vibrations of nonsymmetrically laminated cross-ply L-shaped frames. The proposed theory accounts for the effects of shear deformation, rotary inertia, and the elastic coupling between in-plane bending and longitudinal deformations. The reflection matrices at the boundaries, together with the reflection and transmission matrices at the corner joint of the L-shaped frame are derived. A traveling wave approach is then used to systematically assemble the small-order matrices into a single expression that can be used to efficiently calculate the exact natural frequencies. An expression for evaluating the mode shapes of the frame for general boundary conditions is also given. The application of the wave-based method is illustrated through several numerical examples and the results are validated using independent finite element models. As part of the numerical analysis, the influence of the number of cross-ply layers on the natural frequencies is investigated.

Keywords

Cross-ply laminate, free vibration, L-shaped frame, nonsymmetric, wave analysis

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Introduction

Fiber-reinforced composites (FRCs) represent a class of materials that are widely used to fabricate structural components in sectors such as the aerospace, automotive, robotics, and marine industries. Compared to conventional metals, the main benefit of selecting FRCs are the potential weight savings associated with their high strength-to-weight and stiffness-to-weight ratios. A unique characteristic of composites is that both fiber orientation and stacking sequence can be manipulated to produce specific structural elastic couplings. This property can in turn be exploited to tailor the dynamic characteristics of a structural component to attain a desired reponse.[1,2] Cross-ply laminates in particular have been widely used to produce bending and longitudinal elastic couplings by specifying nonsymmetric [0°/90°] layups.[3,4]

In the literature, an extensive range of studies have been dedicated to analyzing the free vibration characteristics of various composite structural elements. Szekrényes[5] used an analytical approach to model the free vibration behavior of delaminated composite beams. Li et al.[6] investigated the vibrations of composite beams with periodically varying interfaces. Aksencer and Aydogdu[7] studied the effect of a point mass on the flapwise vibrations of a rotating composite beam. Kheladi et al.[8] investigated the free vibration analysis of composite laminated beams reinforced with parabolic fibers using an isogeometric analysis.

A significant amount of work has also specifically focused on the free vibration characteristics of nonsymmetric cross-ply laminated straight beams, curved beams, and plates. Abramovich and Livshits[9] derived the equations of motion for a nonsymmetric cross-ply

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laminated beam and calculated the natural frequencies for different classical boundary conditions. The same authors[10] investigated the free vibration characteristics of symmetric and nonsymmetric cross-ply beams with impregnated piezoelectric layers. Recently, Bachoo[11] used a wave approach to determine expressions for the mode count function and power flow in cross-ply laminated beams, and Khedir and Reddy[12] determined the fundamental natural frequencies of symmetric and nonsymmetric cross-ply arches. Abramovich et al.[13] applied the exact element method to determine the natural frequencies and mode shapes of discontinuous multi-span cross-ply beams, and Jones[14] investigated the effect the number of nonsymmetric cross-ply layers has on the fundamental frequency of laminated plates.

Apart from the study conducted by Abramovich et al.,[13] the dynamic analysis of discontinuous nonsymmetric cross-ply laminated structures has received little attention in literature. There is necessity in analyzing such structures primarily because of their increased use in structural frames,[15] aerospace components,[16] and multimode energy harvesting devices.[17,18] In this paper, the in-plane vibration characteristics of an L-shaped nonsymmetric cross-ply laminated beam is analyzed using a wave approach. The proposed theory includes the effects of shear and rotary inertia, and also accounts for the coupled in-plane flexural and longitudinal deformations. It is demonstrated that upon derivation of the propagation, reflection, and transmission matrices, the wave approach can be used to calculate the exact natural frequencies of the frame by performing a number of simple matrix operations that present no numerical challenges. An expression to determine the mode shapes of the frame with generalized boundary conditions is also derived. The efficacy of the proposed theory is illustrated through the use of several numerical examples. Since no published results are available for the specified problem, independent finite element (FE) models are used to verify the approach. In this work the effect of the number of cross-ply layers on the natural frequencies of the laminated L-beam is also investigated.

Equations of motion and free wave propagation

A typical nonsymmetric cross-ply laminated beam with layup $[0°/90°]_n$ is illustrated in Figure 1(a) and (b). The term n represents the number of times the [0°/90°] stacking sequence is repeated throughout the thickness of the laminate. The layers of lamina that constitute the laminate are assumed to be perfectly bonded. The physical bonds between the interlayers are also assumed to be infinitesimally thin. These underlying assumptions mean that the displacements are continuous across the boundaries of the lamina and hence there is a no-slip condition between adjacent lamina. In practice, particularly in the aerospace and automotive industries, rigorous inspection and tests are conducted to ensure that the layers of a laminate are completely bonded and behave as a single structural unit. Hence, the assumption of a perfectly bonded laminate is often justified and appropriate.

In line with the first-order shear deformation theory the continuous longitudinal displacement is of the form[9]

$$U = u(x,t) + z\psi(x,t) \tag{1}$$

where $u(x,t)$ is the longitudinal displacements of the center line of the beam, $\psi(x,t)$ is the slope due to bending and x, z are the longitudinal and lateral coordinates as shown in Figure 1(b). By definition, the normal and shear strains may be given as

$$\varepsilon_x = \frac{\partial}{\partial x}(u + z\psi) \tag{2a}$$

$$\gamma_{xz} = \psi + \frac{\partial w}{\partial x} \tag{2b}$$

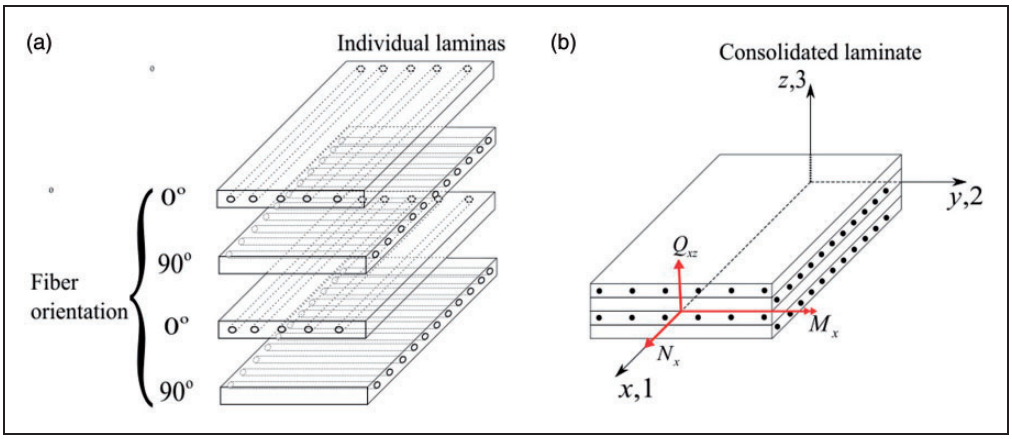


Figure 1. Illustration of a nonsymmetric cross-ply beam: (a) stacking sequence and (b) consolidated laminate with coordinate axes.

respectively, where w is the lateral displacement of the center line of the beam.

The stress–strain relations for each lamina are given as[19]

$$\sigma_x = \bar{Q}_{11}\varepsilon_x = \begin{pmatrix} Q_{11}\cos^4(\theta) + Q_{22}\sin^4(\theta) \\ +2(Q_{12}+2Q_{66})\sin^2(\theta)\cos^2(\theta) \end{pmatrix}\varepsilon_x \tag{3a}$$

$$\tau_{xz} = \bar{Q}_{55}\gamma_{xz} = \left(G_{13}\cos^2(\theta) + G_{23}\sin^2(\theta)\right)\gamma_{xz} \tag{3b}$$

where

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}} \tag{4a}$$

$$Q_{12} = \frac{\nu_{12}E_1}{1-\nu_{12}\nu_{21}} \tag{4b}$$

$$Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}} \tag{4c}$$

$$Q_{66} = G_{12} \tag{4d}$$

$$\nu_{21} = \left(\frac{E_2}{E_1}\right)\nu_{12} \tag{4e}$$

and E_1, E_2, G_{12}, G_{13}, G_{23}, and ν_{12} are the elastic constants of each layer. The subscripts 1, 2, 3 are the principal directions associated with each lamina's orthotropic symmetry. For the case of a nonsymmetric cross-ply laminate, θ is restricted to values of 0° and 90°, depending on the fiber orientation of the lamina. Upon considering equations (3a) to (3b), it is therefore the case that the normal and shear stress distribution are discontinuous at the boundaries between lamina; a characteristic that is in contrast to the continuity associated with the displacements and strains.

Abramovich et al.[9] used the stress–strain relations in equations (3a) to (3b) to formulate the appropriate constitutive equations and derive the following equations of motion for a nonsymmetric cross-ply laminated beam coupled in bending and longitudinal vibrations

$$D_{11}\frac{\partial^2\psi(x,t)}{\partial x^2} + B_{11}\frac{\partial^2 u(x,t)}{\partial x^2} - A_{55}\left(\frac{\partial w(x,t)}{\partial x} + \psi(x,t)\right) - \rho I\ddot{\psi}(x,t) = 0 \tag{5a}$$

$$A_{55}\left(\frac{\partial^2 w(x,t)}{\partial x^2} + \frac{\partial\psi(x,t)}{\partial x}\right) - \rho A\ddot{w}(x,t) = 0 \tag{5b}$$

$$A_{11}\frac{\partial^2 u(x,t)}{\partial x^2} + B_{11}\frac{\partial^2\psi(x,t)}{\partial x^2} - \rho A\ddot{u}(x,t) = 0 \tag{5c}$$

In equations (5a) to (5c), t denotes time, A is the cross-sectional area, I is the second moment of area of the cross-section, and the terms A_{11}, A_{55}, D_{11}, and B_{11} are the longitudinal, shear, bending, and bending–longitudinal coupling rigidities, respectively. The overhead symbol $(\dot{\cdot})$ represents the double differentiation with respect to time.

The expressions for determining the rigidities of a cross-ply laminated beam in terms of the orthotropic elastic constants and number of layers are[11]

$$A_{11} = \frac{E_1(E_1+E_2)h\varepsilon}{2\left(E_1 - E_2\nu_{12}^2\right)} \tag{6a}$$

$$B_{11} = \frac{E_1(E_1-E_2)h^2\varepsilon}{8n\left(E_1 - E_2\nu_{12}^2\right)} \tag{6b}$$

$$D_{11} = \frac{E_1(E_1+E_2)h^3\varepsilon}{24\left(E_1 - E_2\nu_{12}^2\right)} \tag{6c}$$

$$A_{55} = \frac{\kappa h\varepsilon}{2}(G_{13}+G_{23}) \tag{6d}$$

The constitutive equations that relate the internal forces to the displacements are[9]

$$M_x = D_{11}\frac{\partial\psi}{\partial x} + B_{11}\frac{\partial u}{\partial x} \tag{7a}$$

$$N_x = B_{11}\frac{\partial\psi}{\partial x} + A_{11}\frac{\partial u}{\partial x} \tag{7b}$$

$$Q_{xz} = A_{55}\left(\frac{\partial w}{\partial x} + \psi\right) \tag{7c}$$

where (N_x) is the axial force, (M_x) is the bending moment, and (Q_{xz}) is the shear force at any section of the beam (Figure 1(b)).

The solutions satisfying equations (5a) to (5c) are assumed to be harmonic in time and of the form

$$u(x,t) = u_o e^{-ikx}e^{i\omega t} \tag{8a}$$

$$w(x,t) = w_o e^{-ikx}e^{i\omega t} \tag{8b}$$

$$\psi(x,t) = \psi_o e^{-ikx}e^{i\omega t} \tag{8c}$$

where u_o, w_o, ψ_o are the amplitudes of the corresponding displacement, k is the wavenumber, and ω is the circular frequency in radians per second.

Substituting equations (8a) to (8c) into equation (5) leads to the expression

$$\begin{bmatrix} -k^2B_{11} & ikA_{55} & \rho I\omega^2 - k^2D_{11} - A_{55} \\ 0 & \rho A\omega^2 - k^2A_{55} & -ikA_{55} \\ \rho A\omega^2 - k^2A_{11} & 0 & -k^2B_{11} \end{bmatrix} \begin{Bmatrix} u_o \\ w_o \\ \psi_o \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} \tag{9}$$

To determine the nontrivial solution for equation (9) we evaluate the determinant of the above 3×3 matrix to obtain

$$ak^6 + bk^4 + ck^2 + d = 0 \tag{10}$$

where

$$a = A_{11}D_{11} - B_{11}^2 \tag{11a}$$

$$b = \frac{\omega^2\left(\rho AB_{11}^2 - \rho AD_{11}A_{55} - A_{11}(\rho AD_{11} + \rho IA_{55})\right)}{A_{55}} \tag{11b}$$

$$c = \frac{\rho A\omega^2\left(A_{11}\left(\rho I\omega^2 - A_{55}\right) + \omega^2(\rho AD_{11} + \rho IA_{55})\right)}{A_{55}} \tag{11c}$$

$$d = \rho A^2\omega^4\left(1 - \frac{\rho I\omega^2}{A_{55}}\right) \tag{11d}$$

Equation (10) is the dispersion equation of the beam and since it is a polynomial in $\bar{x} = k^2$, it may be rewritten as

$$a\bar{x}^3 + b\bar{x}^2 + c\bar{x} + d = 0 \tag{12}$$

A detailed analysis of the roots of equation (12) is given in Abramovich and Livshits[9] and Bachoo[11] and will not be replicated here. Bachoo[11] and Mei[20] showed that within the frequency range $0 < \omega < \sqrt{A_{55}/(\rho I)}$, the solutions of equation (12) are

$$\bar{x}_1 = \tilde{n} + \tilde{q}\cos(\gamma) \tag{13a}$$

$$\bar{x}_2 = \tilde{n} + \tilde{q}\cos\left(\frac{2\pi}{3} + \gamma\right) \tag{13b}$$

$$\bar{x}_3 = \tilde{n} + \tilde{q}\cos\left(\frac{2\pi}{3} - \gamma\right) \tag{13c}$$

where

$$\tilde{n} = -\frac{b}{3a} \tag{14a}$$

$$\tilde{q} = \frac{2}{3a}\sqrt{b^2 - 3ac} \tag{14b}$$

$$\gamma = \frac{1}{3}\cos^{-1}(-\tilde{F}) \tag{14c}$$

$$\tilde{F} = \frac{4}{a\tilde{q}^3}(a\tilde{n}^3 + b\tilde{n}^2 + c\tilde{n} + d) \tag{14d}$$

The shear rigidity term A_{55} is much greater than the rotary inertia term (ρI) for practical slender beam-like structures, and therefore the ratio $\sqrt{A_{55}/(\rho I)}$ is often greater than $40{,}000\pi$ rad/s (20 kHz). Since the value of $40{,}000\pi$ rad/s (20 kHz) is the well-known upper limit of the audible frequency range, it follows that the solutions presented in equations (13a) to (13c) are valid for most practical structural vibration problems.

In equations (13a) and (13c), $\bar{x}_1$ and $\bar{x}_3$ are both positive-valued roots that lead to two separate pairs of propagating waves

$$k_1 = \sqrt[+]{\bar{x}_1} \tag{15a}$$

$$k_3 = \sqrt[+]{\bar{x}_3} \tag{15b}$$

In equation (13b) $\bar{x}_2$ is a negative-valued root, which leads to a pair of decaying evanescent waves

$$k_2 = \sqrt[+]{|\bar{x}_2|} \tag{16}$$

Therefore, suppressing the time dependence $e^{i\omega t}$ in equations (8a) to (8c), the response of the beam can be expressed as

$$w(x) = b_1e^{-ik_1x} + b_2e^{-k_2x} + b_3e^{-ik_3x} + b_4e^{ik_1x} + b_5e^{k_2x} + b_6e^{ik_3x} \tag{17a}$$

$$u(x) = c_1e^{-ik_1x} + c_2e^{-k_2x} + c_3e^{-ik_3x} + c_4e^{ik_1x} + c_5e^{k_2x} + c_6e^{ik_3x} \tag{17b}$$

$$\psi(x) = d_1e^{-ik_1x} + d_2e^{-k_2x} + d_3e^{-ik_3x} + d_4e^{ik_1x} + d_5e^{k_2x} + d_6e^{ik_3x} \tag{17c}$$

where b_i,c_i,d_i are the amplitudes of the associated wave components and $i = 1,2,...6$.

Using equation (9) the wave amplitudes in equations (17a) to (17c) are related by the expressions

$$\mathbf{b}^{+} = \mathbf{\Pi}^{+}\mathbf{d}^{+} \tag{18a}$$

$$\mathbf{b}^{-} = -\mathbf{\Pi}^{+}\mathbf{d}^{-} \tag{18b}$$

$$\mathbf{c}^{+} = \mathbf{\Pi}^{-}\mathbf{d}^{+} \tag{18c}$$

$$\mathbf{c}^{-} = \mathbf{\Pi}^{-}\mathbf{d}^{-} \tag{18d}$$

where

$$\mathbf{d}^{+} = \left\{ d_1 \quad d_2 \quad d_2 \right\}^{T} \tag{19a}$$

$$\mathbf{d}^{-} = \left\{ d_4 \quad d_5 \quad d_6 \right\}^{T} \tag{19b}$$

$$\mathbf{b}^{+} = \left\{ b_1 \quad b_2 \quad b_3 \right\}^{T} \tag{19c}$$

$$\mathbf{b}^{-} = \left\{ b_4 \quad b_5 \quad b_6 \right\}^{T} \tag{19d}$$

$$\mathbf{c}^{+} = \left\{ c_1 \quad c_2 \quad c_3 \right\}^{T} \tag{19e}$$

$$\mathbf{c}^{-} = \left\{ c_4 \quad c_5 \quad c_6 \right\}^{T} \tag{19f}$$

$$\mathbf{\Pi}^{+} = \mathrm{diag}\begin{bmatrix} \dfrac{\mathrm{i}k_1 A_{55}}{\rho A\omega^2 - k_1^2 A_{55}} & \dfrac{k_2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} \\ \dfrac{\mathrm{i}k_3 A_{55}}{\rho A\omega^2 - k_3^2 A_{55}} & \end{bmatrix} \tag{19g}$$

$$\mathbf{\Pi}^{-} = \mathrm{diag}\begin{bmatrix} \dfrac{k_1^2 B_{11}}{\rho A\omega^2 - k_1^2 A_{11}} & -\dfrac{k_2^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} \\ \dfrac{k_3^2 B_{11}}{\rho A\omega^2 - k_3^2 A_{11}} & \end{bmatrix} \tag{19h}$$

Theoretical analysis of the L-shaped frame

The free vibration characteristics of the L-shaped composite frame shown in Figure 2 are analyzed from a wave standpoint. In practice, the corner joint of the frame is often achieved by using an appropriate adhesive to connect the adjacent members.[21] The adhesive may be directly applied to both members (Figure 3(a)) or through the use of an auxiliary connector (Figure 3(b)).

In the subsequent sections the following are derived: (1) reflection and transmission matrices at the corner joint, (2) the reflection matrices at the boundaries, and (3) the propagation matrices. The principle of wave train phase closure is then used to assemble the matrices and derive an expression for the characteristic equation of the frame.

Reflection and transmission coefficients

Compatibility conditions. The basic dynamic methodology used to assemble the frame and model the corner joint is illustrated in Figure 4. Firstly, the kinematic and kinetic relations that enforce continuity in the displacement fields and compatibility of the axial force, shear force and bending moment at the joint are established. Subsequently, the specific case of a wavefield originating from Beam 1 that leads to reflected and transmitted wavefields on both sides of the joint is considered. The established continuity and compatibility equations are then used to derive the matrices that govern the amplitudes of the reflected and transmitted waves at the junction. The process can then be extended to deal with reflected and transmitted waves across the joint when the incident wave now originates from Beam 2.

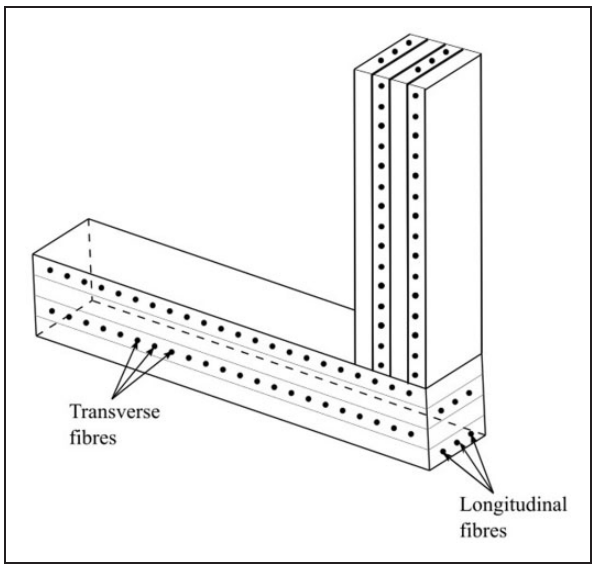


Figure 2. L-shaped nonsymmetric cross-ply laminated frame.

Figure 5 shows the free body diagram of the L-joint together with the associated coordinate systems. At the junction, the transverse, rotational, and longitudinal displacements from the horizontal beam (Beam 1) constrains the motion of the adjacent vertical beam (Beam 2). The conditions required to maintain continuity of the displacement field and compatibility of the internal forces at the junction are

$$u_1 = u_{\mathrm{jn}} \tag{20a}$$

$$\psi_1 = \psi_2 = \psi_{\mathrm{jn}} \tag{20b}$$

$$u_2 = w_{\mathrm{jn}} \tag{20c}$$

$$w_1 - w_{\mathrm{jn}} - \frac{h_2}{2}\psi_{\mathrm{jn}} = 0 \tag{20d}$$

$$w_2 + u_{\mathrm{jn}} + \frac{h_1}{2}\psi_{\mathrm{jn}} = 0 \tag{20e}$$

$$N_{x,2} - Q_{xz,1} = m\ddot{w}_{\mathrm{jn}} \tag{20f}$$

$$N_{x,1} + Q_{xz,2} = -m\ddot{u}_{\mathrm{jn}} \tag{20g}$$

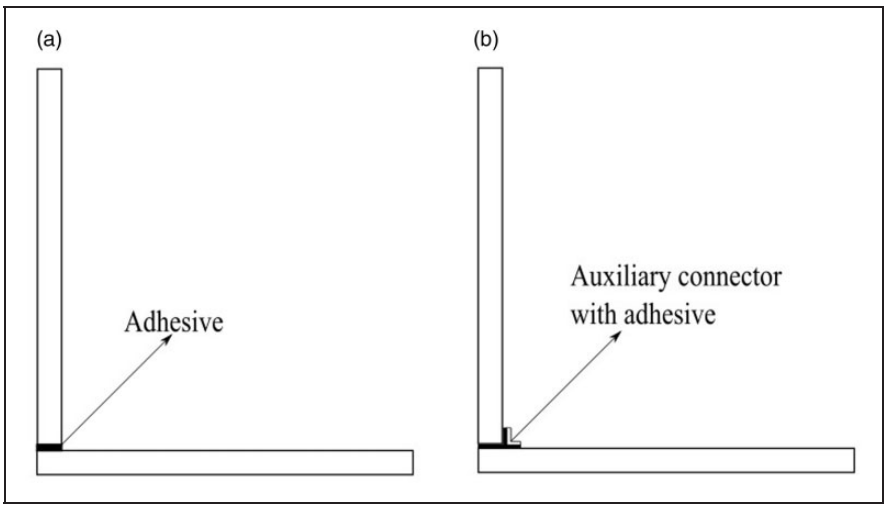


Figure 3. Connection of a corner joint.

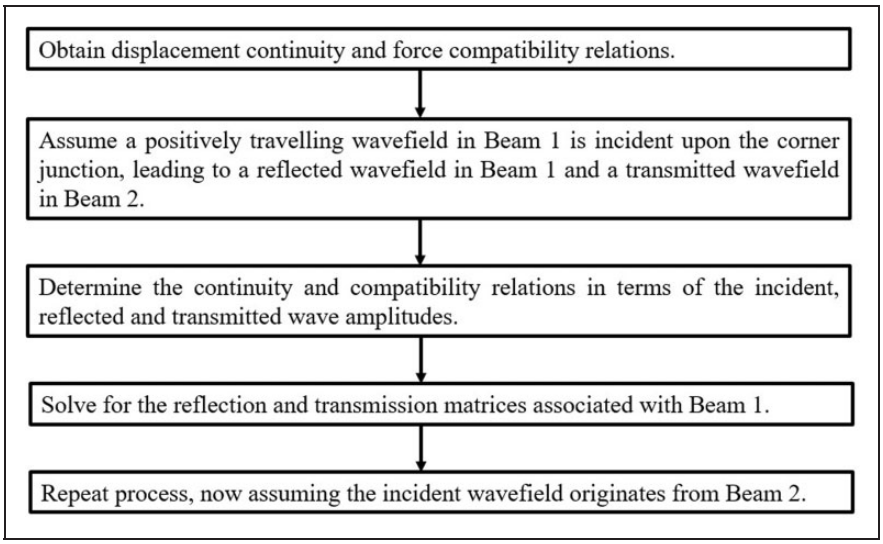


Figure 4. Flowchart illustrating the process to determine the reflection and transmission matrices at the corner joint.

$$M_{x,2} - M_{x,1} - \frac{h_1}{2} Q_{xz,2} - \frac{h_2}{2} Q_{xz,1} = J\ddot{\psi}_{\mathrm{jn}} \tag{20h}$$

The subscripts 1, 2 and jn in equations (20) and Figure 5 indicate that the parameter is associated with either Beam 1, Beam 2 or the junction respectively.

Wave analysis at the corner joint. When a positively travelling wavefield in Beam 1 is incident upon the corner joint, both a reflected wavefield in Beam 1 and a transmitted wavefield in Beam 2 are generated (Figure 6(a)). In alignment with equations (17a) to (17c) the incident and reflected displacement wavefields of Beam 1 are represented as

$$w_{\mathrm{Beam},1}(x) = b_1 \mathrm{e}^{-\mathrm{i}k_1 x} + b_2 \mathrm{e}^{-k_2 x} + b_3 \mathrm{e}^{-\mathrm{i}k_3 x} + b_4 \mathrm{e}^{\mathrm{i}k_1 x} + b_5 \mathrm{e}^{k_2 x} + b_6 \mathrm{e}^{\mathrm{i}k_3 x} \tag{21a}$$

$$u_{\mathrm{Beam},1}(x) = c_1 \mathrm{e}^{-\mathrm{i}k_1 x} + c_2 \mathrm{e}^{-k_2 x} + c_3 \mathrm{e}^{-\mathrm{i}k_3 x} + c_4 \mathrm{e}^{\mathrm{i}k_1 x} + c_5 \mathrm{e}^{k_2 x} + c_6 \mathrm{e}^{\mathrm{i}k_3 x} \tag{21b}$$

$$\psi_{\mathrm{Beam},1}(x) = d_1 \mathrm{e}^{-\mathrm{i}k_1 x} + d_2 \mathrm{e}^{-k_2 x} + d_3 \mathrm{e}^{-\mathrm{i}k_3 x} + d_4 \mathrm{e}^{\mathrm{i}k_1 x} + d_5 \mathrm{e}^{k_2 x} + d_6 \mathrm{e}^{\mathrm{i}k_3 x} \tag{21c}$$

whilst the transmitted wavefield in Beam 2 is represented as

$$w_{\mathrm{Beam},2}(x) = b_1^{+} \mathrm{e}^{-\mathrm{i}k_1 x} + b_2^{+} \mathrm{e}^{-k_2 x} + b_3^{+} \mathrm{e}^{-\mathrm{i}k_3 x} \tag{22a}$$

$$u_{\mathrm{Beam},2}(x) = c_1^{+} \mathrm{e}^{-\mathrm{i}k_1 x} + c_2^{+} \mathrm{e}^{-k_2 x} + c_3^{+} \mathrm{e}^{-\mathrm{i}k_3 x} \tag{22b}$$

$$\psi_{\mathrm{Beam},2}(x) = d_1^{+} \mathrm{e}^{-\mathrm{i}k_1 x} + d_2^{+} \mathrm{e}^{-k_2 x} + d_3^{+} \mathrm{e}^{-\mathrm{i}k_3 x} \tag{22c}$$

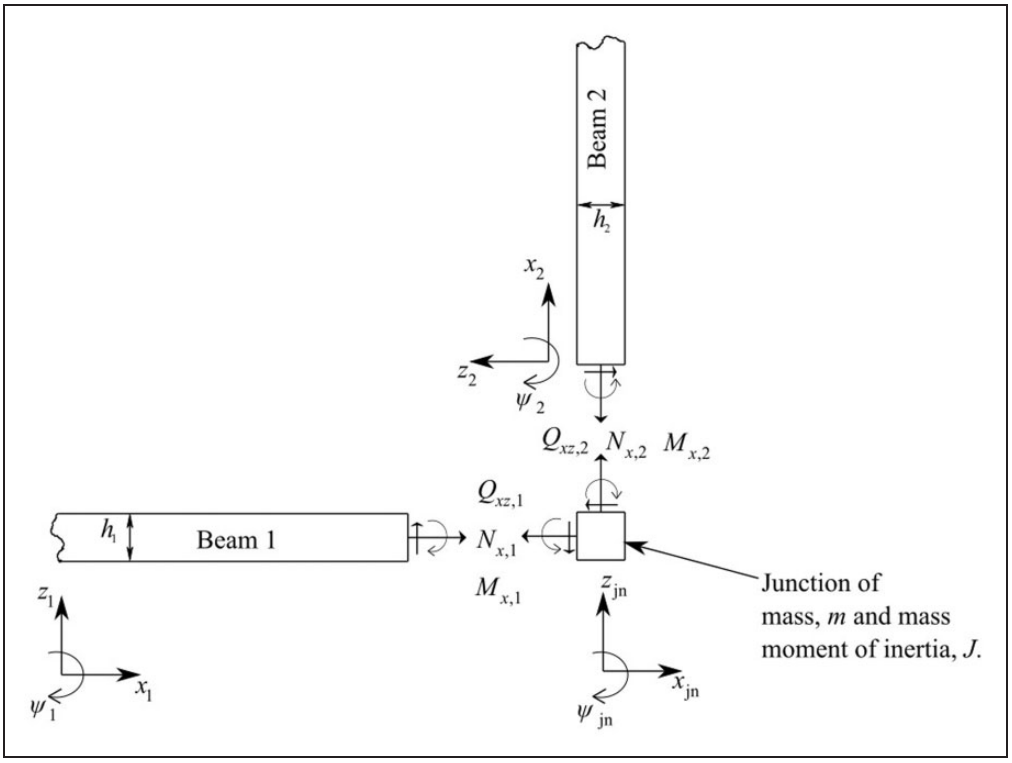


Figure 5. Displacement and force compatibility at the junction of an L-shaped composite planar frame.

Upon substituting equations (21) and (22) into equations equations (20c) to (20e) and using the relations in equations (18a) to (18d) we obtain

$$\mathbf{V}_1^+ . \begin{Bmatrix} d_1^+ \\ d_2^+ \\ d_3^+ \end{Bmatrix} + \mathbf{V}_2^+ . \begin{Bmatrix} d_4 \\ d_5 \\ d_6 \end{Bmatrix} = \mathbf{V}_3^+ . \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} \tag{23}$$

where

$$\mathbf{V}_1^+ = \begin{bmatrix} \frac{h}{2} + \frac{k_1^2 B_{11}}{\rho A\omega^2 - k_1^2 A_{11}} & \frac{h}{2} - \frac{k_2^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & \frac{h}{2} + \frac{k_3^2 B_{11}}{\rho A\omega^2 - k_3^2 A_{11}} \\ \frac{h}{2} + \frac{\mathrm{i}k_1 A_{55}}{\rho A\omega^2 - k_1^2 A_{55}} & \frac{h}{2} + \frac{k_2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & \frac{h}{2} + \frac{\mathrm{i}k_3 A_{55}}{\rho A\omega^2 - k_3^2 A_{55}} \\ -1 & -1 & -1 \end{bmatrix} \tag{24a}$$

$$\mathbf{V}_2^+ = \begin{bmatrix} -\frac{\mathrm{i}k_1 A_{55}}{-\rho A\omega^2 + k_1^2 A_{55}} & \frac{k_2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & -\frac{\mathrm{i}k_3 A_{55}}{-\rho A\omega^2 + k_3^2 A_{55}} \\ \frac{k_1^2 B_{11}}{\rho A\omega^2 - k_1^2 A_{11}} & -\frac{k_2^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & \frac{k_3^2 B_{11}}{\rho A\omega^2 - k_3^2 A_{11}} \\ 1 & 1 & 1 \end{bmatrix} \tag{24b}$$

$$\mathbf{V}_3^+ = \begin{bmatrix} \frac{\mathrm{i}k_1 A_{55}}{\rho A\omega^2 - k_1^2 A_{55}} & \frac{k_2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & \frac{\mathrm{i}k_3 A_{55}}{\rho A\omega^2 - k_3^2 A_{55}} \\ -\frac{k_1^2 B_{11}}{\rho A\omega^2 - k_1^2 A_{11}} & \frac{k_2^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & -\frac{k_3^2 B_{11}}{\rho A\omega^2 - k_3^2 A_{11}} \\ -1 & -1 & -1 \end{bmatrix} \tag{24c}$$

Similarly, substituting equations (21) and (22) into equations (20f) to (20h) leads to

$$\mathbf{V}_6^+ . \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \mathbf{V}_4^+ . \begin{Bmatrix} d_1^+ \\ d_2^+ \\ d_3^+ \end{Bmatrix} + \mathbf{V}_5^+ . \begin{Bmatrix} d_4 \\ d_5 \\ d_6 \end{Bmatrix} \tag{25}$$

where

$$\mathbf{V}_4^+ = \begin{bmatrix} \dfrac{k_1(-k_1 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k_1^2 A_{11}} & -\dfrac{k_2(k_2 m + \rho A)\omega^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & \dfrac{k_3(-k_3 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k_3^2 A_{11}} \\ \dfrac{\rho A\omega^2 A_{55}}{-\rho A\omega^2 + k_1^2 A_{55}} & -\dfrac{\rho A\omega^2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & \dfrac{\rho A\omega^2 A_{55}}{-\rho A\omega^2 + k_3^2 A_{55}} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{26a}$$

$$\mathbf{V}_5^+ = \begin{bmatrix} \dfrac{\rho A\omega^2 A_{55}}{-\rho A\omega^2 + k_1^2 A_{55}} & -\dfrac{\rho A\omega^2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & \dfrac{\rho A\omega^2 A_{55}}{-\rho A\omega^2 + k_3^2 A_{55}} \\ \dfrac{k_1(-k_1 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k_1^2 A_{11}} & -\dfrac{k_2(k_2 m + \rho A)\omega^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & \dfrac{k_3(-k_3 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k_3^2 A_{11}} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \tag{26b}$$

$$\mathbf{V}_6^+ = \begin{bmatrix} \dfrac{\rho A\omega^2 A_{55}}{\rho A\omega^2 - k_1^2 A_{55}} & \dfrac{\rho A\omega^2 A_{55}}{\rho A\omega^2 + k_2^2 A_{55}} & \dfrac{\rho A\omega^2 A_{55}}{\rho A\omega^2 - k_3^2 A_{55}} \\ \dfrac{k_1(k_1 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k_1^2 A_{11}} & \dfrac{k_2(k_2 m - \rho A)\omega^2 B_{11}}{\rho A\omega^2 + k_2^2 A_{11}} & \dfrac{k_3(k_3 m + \mathrm{i}\rho A)\omega^2 B_{11}}{-\rho A\omega^2 + k3^2 A_{11}} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \tag{26c}$$

$$\begin{Bmatrix} a_{31} \\ a_{32} \\ a_{33} \end{Bmatrix} = \begin{Bmatrix} J\omega^2 + \dfrac{\mathrm{i}k_1^3 B_{11}^2}{-\rho A\omega^2 + k_1^2 A_{11}} - \mathrm{i}k_1 D_{11} - \dfrac{1}{2}hA_{55} + \dfrac{hk_1^2 A_{55}^2}{-2\rho A\omega^2 + 2k_1^2 A_{55}} \\ J\omega^2 + \dfrac{k_2^3 B_{11}^2}{\rho A\omega^2 + k_2^2 A_{11}} - k_2 D_{11} - \dfrac{1}{2}hA_{55} + \dfrac{hk_2^2 A_{55}^2}{2\rho A\omega^2 + 2k_2^2 A_{55}} \\ J\omega^2 + \dfrac{\mathrm{i}k_3^3 B_{11}^2}{-\rho A\omega^2 + k_3^2 A_{11}} - \mathrm{i}k_3 D_{11} - \dfrac{1}{2}hA_{55} + \dfrac{hk_3^2 A_{55}^2}{-2\rho A\omega^2 + 2k_3^2 A_{55}} \end{Bmatrix} \tag{26d}$$

$$\begin{Bmatrix} b_{31} \\ b_{32} \\ b_{33} \end{Bmatrix} = \begin{Bmatrix} \dfrac{\mathrm{i}k_1^3 B_{11}^2}{-\rho A\omega^2 + k_1^2 A_{11}} - \mathrm{i}k_1 D_{11} + \dfrac{h\rho A\omega^2 A_{55}}{-2\rho A\omega^2 + 2k_1^2 A_{55}} \\ \dfrac{k_2^3 B_{11}^2}{\rho A\omega^2 + k_2^2 A_{11}} - k_2 D_{11} - \dfrac{h\rho A\omega^2 A_{55}}{2\rho A\omega^2 + 2k_2^2 A_{55}} \\ \dfrac{\mathrm{i}k_3^3 B_{11}^2}{-\rho A\omega^2 + k_3^2 A_{11}} - \mathrm{i}k_3 D_{11} + \dfrac{h\rho A\omega^2 A_{55}}{-2\rho A\omega^2 + 2k_3^2 A_{55}} \end{Bmatrix} \tag{26e}$$

$$\begin{Bmatrix} c_{31} \\ c_{32} \\ c_{33} \end{Bmatrix} = \begin{Bmatrix} \dfrac{\mathrm{i}k_1^3 B_{11}^2}{-\rho A\omega^2 + k_1^2 A_{11}} - \mathrm{i}k_1 D_{11} + \dfrac{h\rho A\omega^2 A_{55}}{2\rho A\omega^2 - 2k_1^2 A_{55}} \\ \dfrac{k_2^3 B_{11}^2}{\rho A\omega^2 + k_2^2 A_{11}} - k_2 D_{11} + \dfrac{h\rho A\omega^2 A_{55}}{2\rho A\omega^2 + 2k_2^2 A_{55}} \\ \dfrac{\mathrm{i}k_3^3 B_{11}^2}{-\rho A\omega^2 + k_3^2 A_{11}} - \mathrm{i}k_3 D_{11} + \dfrac{h\rho A\omega^2 A_{55}}{2\rho A\omega^2 - 2k_3^2 A_{55}} \end{Bmatrix} \tag{26f}$$

The reflection and transmission matrices may now be defined using the expressions

$$\begin{Bmatrix} d_1^+ \\ d_2^+ \\ d_3^+ \end{Bmatrix} = \mathbf{t}_{12}.\begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} \tag{27}$$

and

$$\begin{Bmatrix} d_4 \\ d_5 \\ d_6 \end{Bmatrix} = \mathbf{r}_{11}\begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} \tag{28}$$

where $\mathbf{t}_{12}$ and $\mathbf{r}_{11}$ are the transmission and reflection matrices, respectively. Upon solving equations (23) and (25) the reflection and transmission matrices may be obtained as

$$\mathbf{r}_{11} = \left((\mathbf{V}_1^+)^{-1}.\mathbf{V}_2^+ - (\mathbf{V}_4^+)^{-1}.\mathbf{V}_5^+\right)^{-1} .\left((\mathbf{V}_1^+)^{-1}.\mathbf{V}_3^+ - (\mathbf{V}_4^+)^{-1}.\mathbf{V}_6^+\right) \tag{29}$$

$$\mathbf{t}_{12} = \left((\mathbf{V}_2^+)^{-1}.\mathbf{V}_1^+ - (\mathbf{V}_5^+)^{-1}.\mathbf{V}_4^+\right)^{-1} .\left((\mathbf{V}_2^+)^{-1}.\mathbf{V}_3^+ - (\mathbf{V}_5^+)^{-1}.\mathbf{V}_6^+\right) \tag{30}$$

A completely analogous approach can be used to determine the reflection matrix $\mathbf{r}_{22}$ and transmission matrix $\mathbf{t}_{21}$ associated with an incident wavefield that originates from Beam 2 (Figure 6(b)).

Reflection and propagation matrices

When an incident wavefield impinges on the boundary of the L-frame, a reflected wavefield is generated (Figure 7). The reflection coefficients at the boundary governs the amplitudes of the reflected waves and is determined from the relationship

$$\mathbf{A}^+ = \mathbf{r}.\mathbf{A}^- \tag{31}$$

where $\mathbf{A}^+$ and $\mathbf{A}^-$ are the reflected and incident wavefields respectively and $\mathbf{r}$ is the reflection matrix. The reflection matrix for each end of the L-frame can be obtained from the associated boundary conditions.

In this study, the following classical boundary conditions are considered:

A. Clamped end constraint

$$\psi(x=0) = \mathbf{0} \tag{32a}$$

$$u(x=0) = \mathbf{0} \tag{32b}$$

$$w(x=0) = \mathbf{0} \tag{32c}$$

B. Free end constraint:

$$M_x(x=0) = \mathbf{0} \tag{33a}$$

$$N_x(x=0) = \mathbf{0} \tag{33b}$$

$$Q_{xz}(x=0) = \mathbf{0} \tag{33c}$$

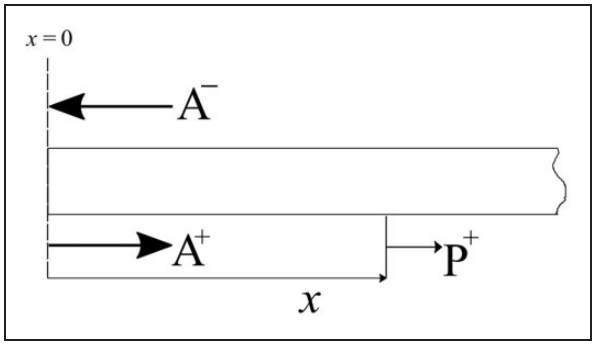


Figure 7. Wave reflection and propagation in a cross-ply laminated beam.

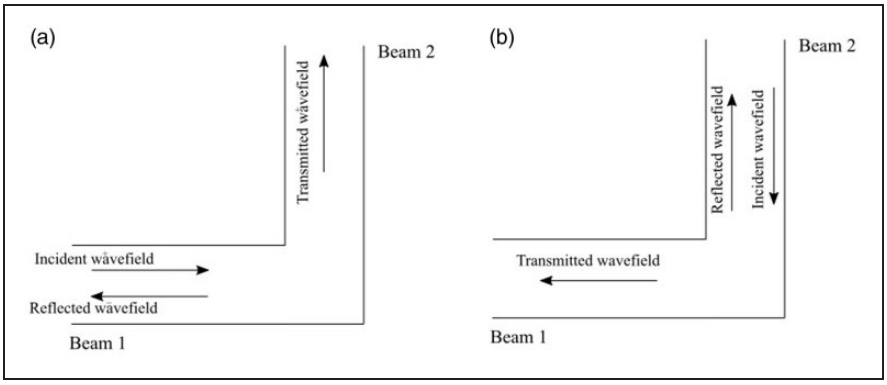


Figure 6. Wavefields at the junction of the corner joint: (a) incident wavefield arriving from Beam 1; (b) incident wavefield arriving from Beam 2.

It follows from equations (31), (32a), (32b), (32c), (17a), (17b), (17c) the reflection matrix at the clamped end is

$$\mathbf{r}_f = \begin{bmatrix} \mathrm{i}k_1 B_{11} + \frac{\mathrm{i}k_1{}^3 A_{11} B_{11}}{\rho A\omega^2 - k_1{}^2 A_{11}} & k_2 B_{11} - \frac{k_2{}^3 A_{11} B_{11}}{\rho A\omega^2 + k_2{}^2 A_{11}} & \mathrm{i}k_3 B_{11} + \frac{\mathrm{i}k_3{}^3 A_{11} B_{11}}{\rho A\omega^2 - k_3{}^2 A_{11}} \\ \frac{\mathrm{i}k_1{}^3 B_{11}^2}{\rho A\omega^2 - k_1{}^2 A_{11}} + \mathrm{i}k_1 D_{11} & -\frac{k_2{}^3 B_{11}^2}{\rho A\omega^2 + k_2{}^2 A_{11}} + k_2 D_{11} & \frac{\mathrm{i}k_3{}^3 B_{11}^2}{\rho A\omega^2 - k_3{}^2 A_{11}} + \mathrm{i}k_3 D_{11} \\ -A_{55}\left(1 + \frac{k_1{}^2 A_{55}}{\rho A\omega^2 - k_1{}^2 A_{55}}\right) & -A_{55}\left(1 - \frac{k_2{}^2 A_{55}}{\rho A\omega^2 + k_2{}^2 A_{55}}\right) & -A_{55}\left(1 + \frac{k_3{}^2 A_{55}}{\rho A\omega^2 - k_3{}^2 A_{55}}\right) \end{bmatrix}^{-1} \times \begin{bmatrix} \mathrm{i}k_1 B_{11} + \frac{\mathrm{i}k_1{}^3 A_{11} B_{11}}{\rho A\omega^2 - k_1{}^2 A_{11}} & k_2 B_{11} - \frac{k_2{}^3 A_{11} B_{11}}{\rho A\omega^2 + k_2{}^2 A_{11}} & \mathrm{i}k_3 B_{11} + \frac{\mathrm{i}k_3{}^3 A_{11} B_{11}}{\rho A\omega^2 - k_3{}^2 A_{11}} \\ \frac{\mathrm{i}k_1{}^3 B_{11}^2}{\rho A\omega^2 - k_1{}^2 A_{11}} + \mathrm{i}k_1 D_{11} & -\frac{k_2{}^3 B_{11}^2}{\rho A\omega^2 + k_2{}^2 A_{11}} + k_2 D_{11} & \frac{\mathrm{i}k_3{}^3 B_{11}^2}{\rho A\omega^2 - k_3{}^2 A_{11}} + \mathrm{i}k_3 D_{11} \\ A_{55}\left(1 - \frac{k_1{}^2 A_{55}}{-\rho A\omega^2 + k_1{}^2 A_{55}}\right) & A_{55}\left(1 - \frac{k_2{}^2 A_{55}}{\rho A\omega^2 + k_2{}^2 A_{55}}\right) & A_{55}\left(1 - \frac{k_3{}^2 A_{55}}{-\rho A\omega^2 + k_3{}^2 A_{55}}\right) \end{bmatrix} \tag{35}$$

$$\mathbf{r}_c = \begin{bmatrix} -\frac{\mathrm{i}k_1 A_{55}}{\rho A\omega^2 - k_1{}^2 A_{55}} & -\frac{k_2 A_{55}}{\rho A\omega^2 + k_2{}^2 A_{55}} & -\frac{\mathrm{i}k_3 A_{55}}{\rho A\omega^2 - k_3{}^2 A_{55}} \\ -1 & -1 & -1 \\ -\frac{k_1{}^2 B_{11}}{\rho A\omega^2 - k_1{}^2 A_{11}} & \frac{k_2{}^2 B_{11}}{\rho A\omega^2 + k_2{}^2 A_{11}} & -\frac{k_3{}^2 B_{11}}{\rho A\omega^2 - k_3{}^2 A_{11}} \end{bmatrix}^{-1} \times \begin{bmatrix} \frac{\mathrm{i}k_1 A_{55}}{-\rho A\omega^2 + k_1{}^2 A_{55}} & -\frac{k_2 A_{55}}{\rho A\omega^2 + k_2{}^2 A_{55}} & \frac{\mathrm{i}k_3 A_{55}}{-\rho A\omega^2 + k_3{}^2 A_{55}} \\ 1 & 1 & 1 \\ \frac{k_1{}^2 B_{11}}{\rho A\omega^2 - k_1{}^2 A_{11}} & -\frac{k_2{}^2 B_{11}}{\rho A\omega^2 + k_2{}^2 A_{11}} & \frac{k_3{}^2 B_{11}}{\rho A\omega^2 - k_3{}^2 A_{11}} \end{bmatrix} \tag{34}$$

Similarly, from equations (31), (33a), (33b), (33c), (17a), (17b), (17c) the reflection matrix at the free end is

The propagation matrix governs the amplitudes of the wavefield as it propagates through a continuous section of the beam. For instance, the wavefield $\mathbf{P}^+$ located at a distance x from the wavefield $\mathbf{A}+$ originating at the boundary is given as

$$\mathbf{P}^+ = \mathbf{f}(x)\mathbf{A}^+ \tag{36}$$

where

$$\boldsymbol{f}(x) = \begin{bmatrix} \mathrm{e}^{-\mathrm{i}k_1 x} & 0 & 0 \\ 0 & \mathrm{e}^{-k_2 x} & 0 \\ 0 & 0 & \mathrm{e}^{-\mathrm{i}k_3 x} \end{bmatrix} \tag{37}$$

is the propagation matrix.

Determination of natural frequencies

The previous derivation of the reflection, transmission and propagation matrices for the L-shaped frame allows for the calculation of the natural frequencies using a systematic approach. The circuit that a wavefield travels through in the L-shaped frame is illustrated in Figure 8 and is governed by the equations

$$\mathbf{C}_1^+ = \mathbf{f}(L_1)\mathbf{A}^+ \tag{38a}$$

$$\mathbf{C}_2^+ = \mathbf{t}_{12}\mathbf{C}_1^+ + \mathbf{r}_{22}\mathbf{C}_2^- \tag{38b}$$

$$\mathbf{B}^+ = \mathbf{f}(L_2)\mathbf{C}_2^+ \tag{38c}$$

$$\mathbf{B}^- = \mathbf{r}_\mathrm{B}\mathbf{B}^+ \tag{38d}$$

$$\mathbf{C}_2^- = \mathbf{f}(L_2)\mathbf{B}^- \tag{38e}$$

$$\mathbf{C}_1^- = \mathbf{t}_{21}\mathbf{C}_2^- + \mathbf{r}_{11}\mathbf{C}_1^+ \tag{38f}$$

$$\mathbf{A}^- = \mathbf{f}(L_1)\mathbf{C}_1^- \tag{38g}$$

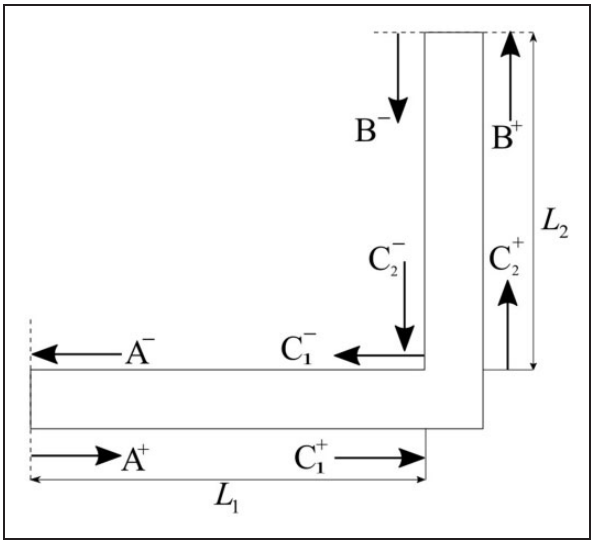


Figure 8. Wave transmission circuit in the L-shaped frame.

$$\mathbf{A}^{+} = \mathbf{r}_{\mathrm{A}}\mathbf{A}^{-} \tag{38h}$$

Upon solving equations (38a) to (38h) for $\mathbf{C}_2^{+}$ we obtain the characteristic equation

$$\begin{bmatrix} [\mathbf{I} - \mathbf{r}_{11}.\mathbf{f}(L_1).\mathbf{r}_{\mathrm{A}}.\mathbf{f}(L_1)].[\mathbf{t}_{12}.\mathbf{f}(L_1).\mathbf{r}_{\mathrm{A}}.\mathbf{f}(L_1)]^{-1} \\ .[\mathbf{I} - \mathbf{r}_{22}.\mathbf{f}(L_2).\mathbf{r}_{\mathrm{B}}.\mathbf{f}(L_2)] - [\mathbf{t}_{21}.\mathbf{f}(L_2).\mathbf{r}_{\mathrm{B}}.\mathbf{f}(L_2)] \end{bmatrix} \mathbf{C}_2^{+} = 0 \tag{39}$$

To determine the nontrivial solutions for equation (39) and hence the natural frequencies of the frame, the expression

$$\left| \begin{matrix} [\mathbf{I} - \mathbf{r}_{11}.\mathbf{f}(L_1).\mathbf{r}_{\mathrm{A}}.\mathbf{f}(L_1)].[\mathbf{t}_{12}.\mathbf{f}(L_1).\mathbf{r}_{\mathrm{A}}.\mathbf{f}(L_1)]^{-1} \\ .[\mathbf{I} - \mathbf{r}_{22}.\mathbf{f}(L_2).\mathbf{r}_{\mathrm{B}}.\mathbf{f}(L_2)] - [\mathbf{t}_{21}.\mathbf{f}(L_2).\mathbf{r}_{\mathrm{B}}.\mathbf{f}(L_2)] \end{matrix} \right| = 0 \tag{40}$$

must be solved. Equation (40) consists of a number of simple sequential matrix operations that can only be evaluated using numerical techniques. Since the previously derived reflection, transmission and propagation matrices are of small order (three); algorithms to determine the numerical solutions of equation (40) can be efficiently executed.

The process of combining the propagation, reflection and transmission matrices to obtain the characteristic equation is referred to as the principle of wave train phase closure.[22] The approach to obtain equation (39) indicates that upon deriving the relevant matrices, the principle of wave train phase closure can easily be extended to deal with planar frames having multiple members. Therefore, the method presented herein can easily be extended to determine the natural frequencies of more geometrically complex nonsymmetrically laminated composite frames.

Mode shapes

The mode shape corresponding to each natural frequency can be obtained by satisfying the associated boundary conditions described in an earlier section (equation (32) or (33)) and the compatibility relations described in the section "Compatibility conditions" (equations (20a) to (20h)). Each boundary condition leads to three separate linear algebraic equations whilst the compatibility relations lead to six separate equations. All twelve equations can be arranged into the matrix form

$$\mathbf{m}.\vec{\mathbf{d}} = 0 \tag{41}$$

where

$$\mathbf{m} = \begin{bmatrix} m_{1,1} & m_{1,2} & \cdots & m_{1,12} \\ m_{2,1} & m_{2,2} & \cdots & m_{2,12} \\ \vdots & \vdots & \ddots & \vdots \\ m_{12,1} & \cdots & \cdots & m_{12,12} \end{bmatrix} \tag{42a}$$

is a 12 × 12 matrix and

$$\vec{\mathbf{d}} = \begin{Bmatrix} d_1 \\ \vdots \\ \vdots \\ d_6^{+} \end{Bmatrix} \tag{43b}$$

is the displacement vector of length 12 × 1.

Upon eliminating an arbitrary row of the 12 × 12 matrix and solving for eleven of the wave amplitudes in terms of an arbitrarily selected wave component, the relative displacements of the mode at the particular natural frequency may be obtained. For instance, if row 12 is deleted and the remaining components are solved for in terms of d_6^{+}, then equation (41) becomes

$$\begin{Bmatrix} d_1 \\ \vdots \\ \vdots \\ d_5^{+} \end{Bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & \cdots & m_{1,11} \\ m_{2,1} & m_{2,2} & \cdots & m_{2,11} \\ \vdots & \vdots & \ddots & \vdots \\ m_{11,1} & \cdots & \cdots & m_{11,11} \end{bmatrix}^{-1} \cdot \begin{Bmatrix} -m_{1,12} \\ -m_{2,12} \\ \vdots \\ -m_{11,12} \end{Bmatrix} d_6^{+} \tag{44}$$

Numerical simulations and validation

In this section numerical results are presented for an L-shaped cross-ply laminated composite frame having different boundary conditions and nonsymmetric stacking sequences. The frame material is AS/3501-6 graphite-epoxy and the base properties as obtained from Abramovich and Livshits[9] are given in Table 1. The shear correction factor (κ) is taken as 5/6 for all numerical simulations and is similar to that used in a wide range of literature.[9–11,13,20] The horizontal and vertical arms of the frame have the same dimensions, $h_1 = h_2 = 0.001$ m, $\varepsilon_1 = \varepsilon_2 = 5h$, and $L_1 = L_2 = 100h$.

Table 1. Material properties of AS/3501-6 graphite-epoxy.[9]

Beam properties	ρ	E_1	E_2	G_{12}	G_{13}	G_{23}	ν_{12}	κ
	(kg/m^3)	(N/m^2)	(N/m^2)	(N/m^2)	(N/m^2)	(N/m^2)		
	1520	145×10^9	9.6×10^9	4.1×10^9	4.1×10^9	3.4×10^9	0.3	5/6

The first set of simulations are done for an L-shaped frame with stacking sequence $[0°/90°]_1$. The ends conditions considered are clamped–free and clamped–clamped. The exact natural frequencies are determined from the characteristic equation (40) using an in-house program written in Matlab 9.6. Figure 9 shows a typical plot of the magnitude of the characteristic equation versus frequency. It can be observed that the points on the curve intersecting with the horizontal axis correspond to individual natural frequencies. Table 2 shows the results of the first four natural frequencies for both sets of boundary conditions.

Since there are no available published results for comparison, the natural frequencies obtained using equation (40) are compared with values obtained from an independently generated FE model. The FE simulation is done using ANSYS APDL 18.0 and the material and geometric properties of the frame are the same as those previously mentioned. The FE model uses 100,000 SHELL181 elements with smart size meshing and the elements are refined at the boundaries and joints. To ensure that the accuracy of the FE simulation is maintained over the entire frequency range analyzed, the number of elements per wavelength exceeds the minimum requirement of six.[23]

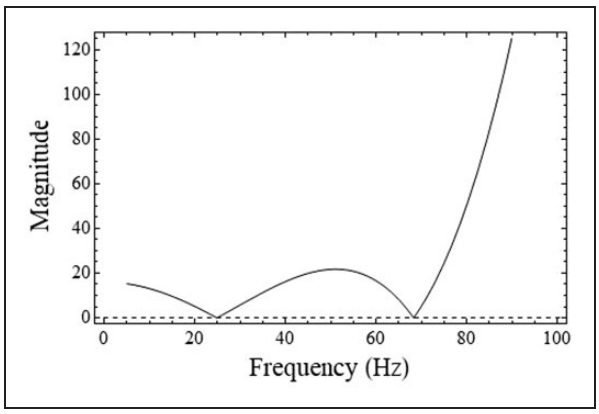


Figure 9. Magnitude of the determinant of the characteristic equation (equation (40)) for a clamped–free beam.

The results of the natural frequencies from the ANSYS model are also presented in Table 2. The deviation between the exact natural frequencies compared with the ANSYS model is always within ±1% for any given mode across both sets of boundary conditions; thereby indicating good agreement. The first four in-plane bending mode shapes for the clamped–free and clamped–clamped beam are determined using equation (41) and shown in Figures 10 and 11, respectively. Although not shown, the mode shapes are also similar to that determined by the ANSYS FE model.

In the second set of simulations the variation of the natural frequencies with stacking sequence is investigated. The number of layers is increased whilst the thickness h of each beam is kept constant. Table 2 shows the natural frequencies for the L-shaped frames having stacking sequences $[0°/90°]_2$ and $[0°/90°]_3$ respectively for both the clamped–free and clamped–clamped boundary conditions. Similar to the $[0°/90°]_1$ layup, the natural frequencies are determined using both equation (40) and independent ANSYS FE models. The validity of equation (40) and by extension the wave approach is once again demonstrated as the exact natural frequencies are always within a ±1% range of the ANSYS result for any given mode and boundary condition.

From Table 2, it is observed that as the number of layers of nonsymmetric cross-ply laminates within a given thickness increases, the natural frequencies also increase. Comparing the natural frequency values as we move from a $[0°/90°]_1$ stacking sequence to a $[0°/90°]_2$ stacking sequence, the natural frequencies of all four modes increase by approximately 42% for both the clamped–free and clamped–clamped cases. Upon performing similar comparisons between the $[0°/90°]_1$ and $[0°/90°]_3$ stacking sequences it can be seen that the natural frequencies of all four modes increase by approximately 48%. The reason for the increase in natural frequency can be explained by considering equations (6a), (6c), and (6d), where it can be

Table 2. Natural frequencies of nonsymmetric cross-ply laminated L-shaped frames.

	$[0°/90°]_1$			$[0°/90°]_2$			$[0°/90°]_3$		
Mode number	Equation (40)	ANSYS	Percent difference	Equation (40)	ANSYS	Percent diffe-rence	Equation (40)	ANSYS	Percent difference
Clamped–free natural frequencies (Hz)									
1	24.89	25.02	−0.52	35.34	35.58	−0.68	36.94	37.21	−0.73
2	68.29	68.08	0.31	96.86	96.77	0.09	101.25	101.22	0.03
3	335.25	334.86	0.12	474.62	476.48	−0.39	495.81	498.68	−0.58
4	492.82	492.98	−0.03	698.85	698.47	0.05	730.58	730.35	0.03
Clamped–clamped natural frequencies (Hz)									
1	327.80	327.76	0.01	463.98	465.98	−0.43	484.70	487.64	−0.61
2	477.83	477.13	0.15	677.08	675.88	0.18	707.66	706.82	0.12
3	1059.87	1059.54	0.03	1497.25	1502.34	−0.34	1563.52	1572.31	−0.56
4	1313.08	1310.76	0.18	1855.35	1849.90	0.29	1938.16	1934.70	0.18

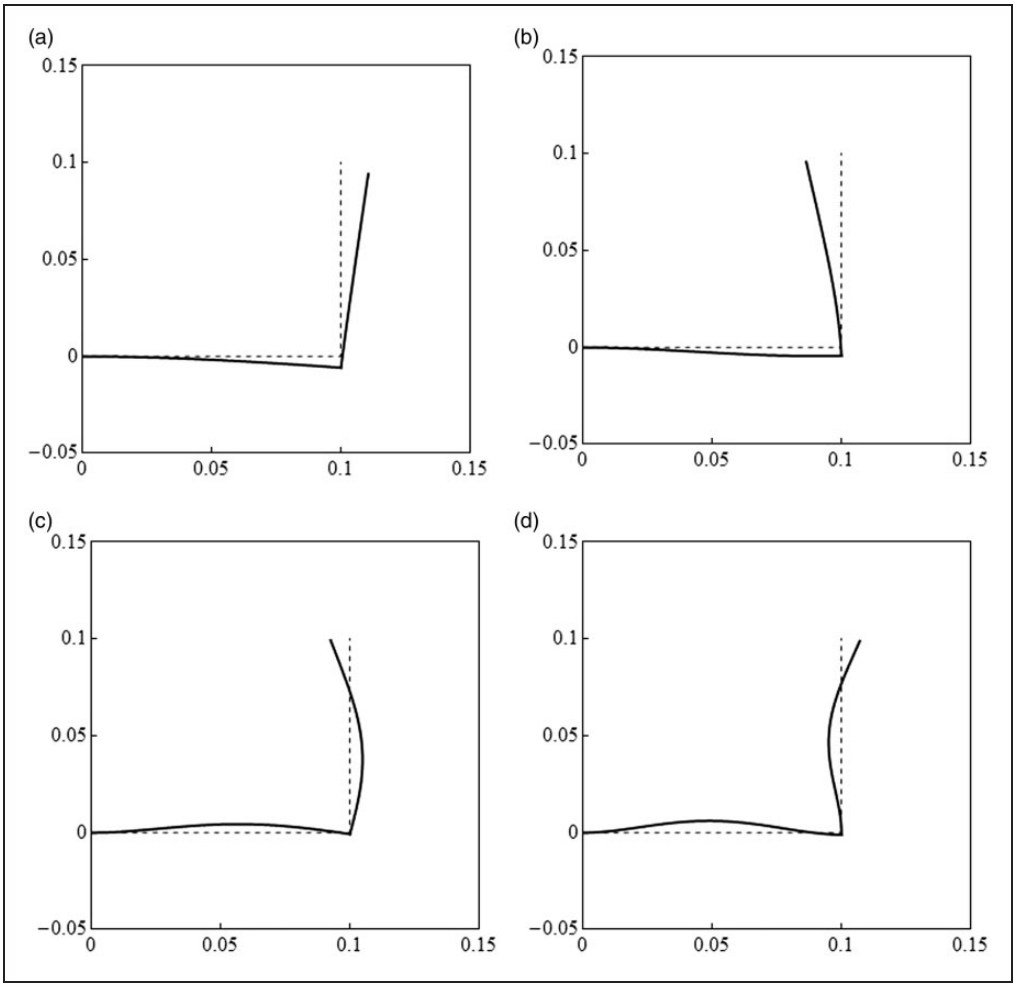


Figure 10. Lateral mode shapes of the clamped–free $[0°/90°]_1$ L-shaped frame: (a) first mode; (b) second mode; (c) third mode; (d) fourth mode. Lateral deflection of frame (—) and undeformed frame (– – –).

observed that neither the bending rigidity (D_{11}), longitudinal rigidity (A_{11}) or shear rigidity (A_{55}) are influenced by the number of layers (if the beam thickness is kept constant). Considering equation (6b), however, one observes that the bending–longitudinal coupling rigidity (B_{11}) decreases with an increase in the number of layers. It must therefore be the case that the reduction in bending–longitudinal coupling associated with an increase in the number of layers is responsible for the increase in natural frequencies.

The final set of numerical simulations will be conducted for an isotropic L-shaped steel frame that has been studied by Lin.[24] It should be noted that the isotropic L-shaped frame represents a simplified case of the more complex cross-ply nonsymmetric beam modeled in this study. Therefore, the final simulations are undertaken to further emphasize the validity of the proposed theory.

From equations (5a) to (5c), it can be shown for an isotropic frame the rigidities are such that

$$D_{11} = EI \tag{45a}$$

$$A_{11} = EA \tag{45b}$$

$$A_{55} = \kappa AG \tag{45c}$$

$$B_{11} = 0 \tag{45d}$$

where E is the Young's modulus, G is the shear modulus, A remains the cross-sectional area, and I is still the second moment of area. The term $B_{11} = 0$ indicates the well-known fact that there is no coupling between the lateral and longitudinal deformations in a *straight continuous uniform* isotropic beam.[19] For the steel frame considered, the horizontal and vertical arms of the frame have the same length $L_1 = L_2 = 0.5$ m and square cross section, (0.0127×0.0127) m^2. The Young's modulus of the frame is $E = 206 \times 10^9$ Pa, Poisson's ratio is $\nu = 0.29$, and the density is $\rho = 7800$ kg/m^3.

Table 3 shows the natural frequencies of the first five modes obtained from equation (40) when the steel frame has clamped–free end conditions. The results obtained from an Ansys APDL simulation are also included in Table 3. The agreement

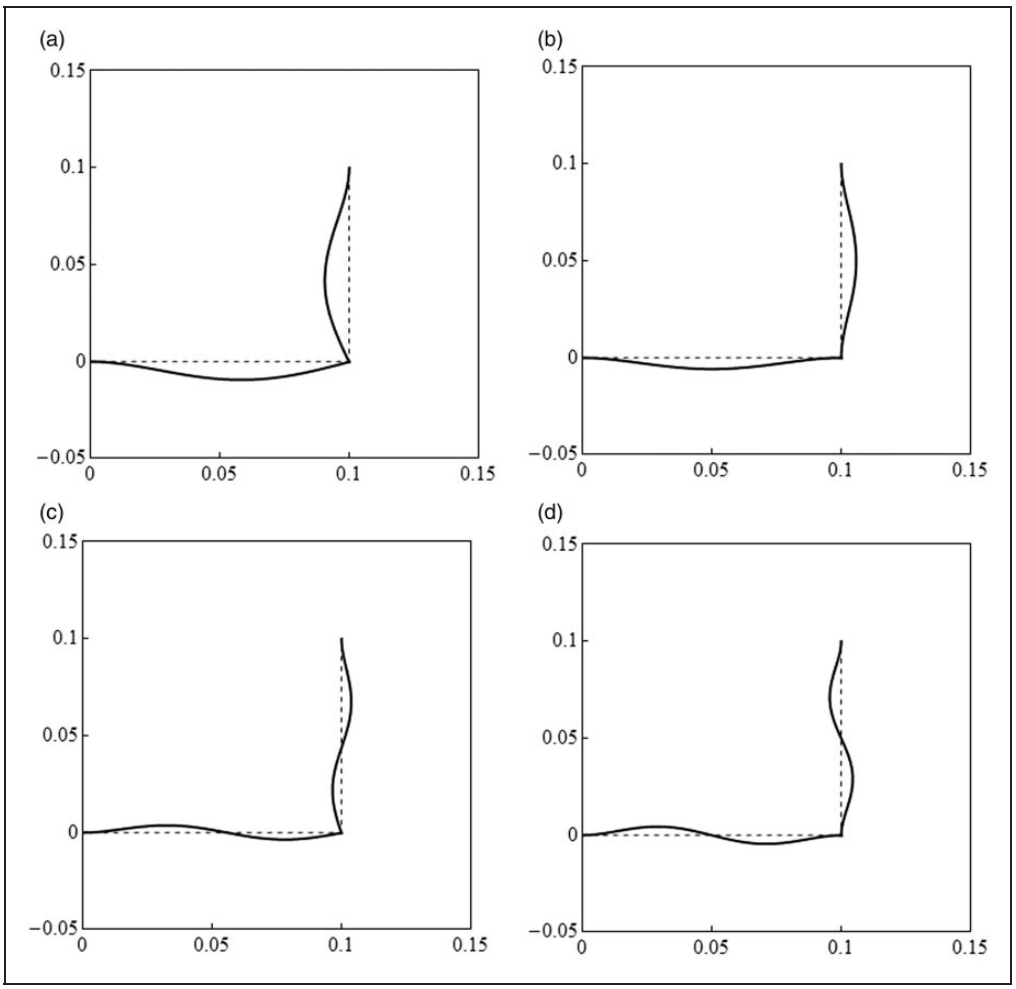


Figure 11. Lateral mode shapes of the clamped–clamped $[0°/90°]_1$ L-shaped frame: (a) first mode; (b) second mode; (c) third mode; (d) fourth mode. Lateral deflection of frame (—) and undeformed frame (– – –).

Table 3. Natural frequencies of an isotropic L-shaped steel frame with clamped–free end conditions.

	Natural frequencies (Hz)					
	Present		Ref.[24]			
Mode number	Equation (40)	ANSYS	Theoretical model	Experimental	Percent difference	Percent difference between equation (40) and Ref. [24] (Experimental)
1	13.74	13.7	14.2	14	1.4	−1.9
2	38.08	37.3	38.3	38	0.8	0.2
3	183.8	183.9	188.9	184	2.7	−0.1
4	275.3	269.0	276.6	269	2.8	2.3
5	584.80	585.8	603.2	583	3.5	−3.1

between equation (40) and the FE simulations are in close agreement for all modes. Lin and Ro[24] studied the same frame theoretically using a hybrid analytical–numerical method and also presented experimentally obtained data. The results from Lin and Ro[24] are presented in Table 3. It can be observed that both the analytical solution (equation (40)) and the Ansys APDL model agree well with the theoretical and experimental data obtained by Lin and Ro.[24] Upon comparing the analytical results of the present work (equation (40)) with the experimental results, it is observed that the magnitude of the percentage deviation ranges between 0% and 3.1%. Lin and Ro[24] also indicated that deviations of similar magnitudes were observed when comparing the results of their own theoretical model with the experimentally obtained data.

Conclusions

In this work, the free vibration characteristics of a nonsymmetric cross-ply laminated L-shaped frame is studied using a travelling wave approach that includes the effects and shear deformation and rotary inertia. Unlike symmetric laminates and isotropic materials, nonsymmetric cross-ply laminates are materially coupled in bending and longitudinal deformations. Consequently, the freely propagating wavefield in a nonsymmetric cross-ply laminated beam constitutes of two propagating (k_1 and k_3) waves and a decaying or evanescent wave (k_2). The L-shaped frame has three main discontinuities; the corner joint and the two boundaries of the frame. In the preceding sections, third-order reflection and transmission matrices at the corner joint are derived by considering the continuity and compatibility conditions. Similarly, third-order reflection matrices at clamped and free ends are also derived using the appropriate boundary conditions. The wave train phase closure principle is then used to combine the propagation, reflection and transmission matrices to produce the characteristic equation of the frame. Solving the characteristic equation requires that a number of simple matrix operations be performed and since the order of the matrices are small, the exact natural frequencies are determined efficiently without numerical challenges.

Application of the proposed wave based theory is illustrated using several numerical examples. The natural frequencies and mode shapes for an equal angle $[0^o/90^o]_1$ L-shaped frame is determined for clamped–free and clamped–clamped end conditions. The analytically determined natural frequencies are compared with those obtained from an independently generated FE model using the commercial software ANSYS. The analytical and numerical results are in good agreement with the differences always being within a $\pm 1\%$ range.

As part of the numerical analysis, the influence of the number of layers on the natural frequencies is also investigated. The bending–longitudinal coupling rigidity is observed to have an inverse relation with the number of layers and consequently decays rapidly as layers are increased. Numerical simulations indicate that this decrease in coupling rigidity leads to a 42% increase across all natural frequencies as the beam transitions from a $[0^\circ/90^\circ]_1$ layup to a $[0^\circ/90^\circ]_2$ layup, regardless of boundary conditions. Similarly, the natural frequencies of a $[0^\circ/90^\circ]_3$ laminated frame are 48% higher than the $[0^\circ/90^\circ]_1$ frame across all modes. The ability to accurately determine and assess the degree to which the natural frequencies and mode shapes of the L-shaped composite beam can be manipulated is a useful tool for engineers seeking to optimize the response of such structural elements.

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Appendix

Notation

a, b, c, d	coefficients of dispersion equation
A	cross-sectional area
A_{11}	longitudinal rigidity
A_{55}	shear rigidity
b_i, c_i, d_i	amplitude of wave components where $i = 1,2,\ldots 6$
B_{11}	bending–longitudinal coupling rigidity
D_{11}	bending rigidity
E	Young's modulus of an isotropic material
E_1	elastic modulus in the principal direction parallel to the fiber
E_2	elastic modulus in the principal direction transverse to the fiber
$\mathbf{f}$	propagation matrix
$\tilde{F}$	variable used in solving the dispersion equation
G	shear modulus of an isotropic material
G_{ij}	shear modulus of elasticity acting in the i direction on a plane with a normal in the j direction; $i = 1,2$ and $j = 2,3$
h	beam thickness
h_1	thickness of beam 1
h_2	thickness of beam 2
I	second moment of area
$\mathbf{I}$	3×3 identity matrix
k	general wavenumber
k_i	propagating wavenumbers, $i = 1,3$
k_2	decaying or evanescent wavenumber
L	beam length
L_1	length of Beam 1
L_2	length of Beam 2
$m_{i,j}$	coefficients of the modal amplitudes; $i, j = 1,2,3$
M_x	internal bending moment
n	number of layers
$\tilde{n}$	variable used in solving the dispersion equation
N_x	internal axial force
Q_{xz}	internal shear force
Q_{ij}	reduced stiffness terms, $i = 1,2,6$ $j = 1,2,6$
$\tilde{q}$	variable used in solving the dispersion equation
$\mathbf{r}_f$, $\mathbf{r}_c$	free and clamped reflection matrix respectively
$\mathbf{r}_{11}$	reflection matrix at corner joint (wavefields reflected in Beam 1)
$\mathbf{r}_{22}$	reflection matrix at corner joint (wavefields reflected in Beam 2)
t	time
$\mathbf{t}_{12}$	transmission matrix at corner joint (wavefields transmitted from Beam 1 to Beam 2)
$\mathbf{t}_{21}$	transmission matrix at corner joint (wavefields transmitted from Beam 2 to Beam 1)
u, w, ψ	displacements due to longitudinal, bending and torsional rotation respectively
U	longitudinal displacement at any distance from the center line of the beam
x, y, z	coordinates
$\bar{x}$	variable in dispersion equation
$\bar{x}_1$, $\bar{x}_2$, $\bar{x}_3$	roots of cubic dispersion equation
ε	beam width
ε_x	longitudinal strain
ε_1	width of Beam 1
ε_2	width of Beam 2
γ	variable used in solving the dispersion equation
γ_{xz}	shear strain
κ	shear correction factor
ν	Poisson's ratio of an isotropic material
ν_{ij}	Poisson's ratio representing contraction in the j direction when a tensile axial load is applied in the i direction, $i, j = 1,2$ and $i \neq j$.
ω	radian frequency
$\mathbf{\Pi}^+, \mathbf{\Pi}^-$	diagonal matrix relating the component wave vector amplitudes
ρ	material density
σ	longitudinal stress
θ	fiber orientation
τ_{xz}	shear stress